6.S892 Advanced Topics in Power Electronics Project Report

Integrated multi-winding transformers for decoupled MAB power flow: a case study of three-winding transformers in TAB

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Research Objectives

- ✓ Intuitively derive the appropriate transformer design.
- ✓ Perform Finite-Element Modeling to validate the proposed design.
- ✓ Verify system performance through simulations.

Background, Problem Statement and Preliminaries

The dual-active-bridge (DAB) converter, as equivalent circuit shown in Fig. 1, featuring a high-frequency transformer structure, together with the symmetrical H-bridges configuration on both the primary and secondary sides, provides a natural potential for extending the system to multiple ports by incorporating additional windings on the same transformer. This leads to the development of multi-active-bridge (MAB) converters. One equivalent circuit model of MAB can be found in Fig. 2 [2]. However, a key challenge in such multiport configurations lies in effectively decoupling the power flow among the different ports.

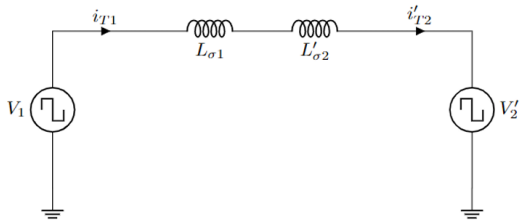


Fig. 1: Equivalent model of a DAB converter, assuming significantly higher magnetizing inductance.

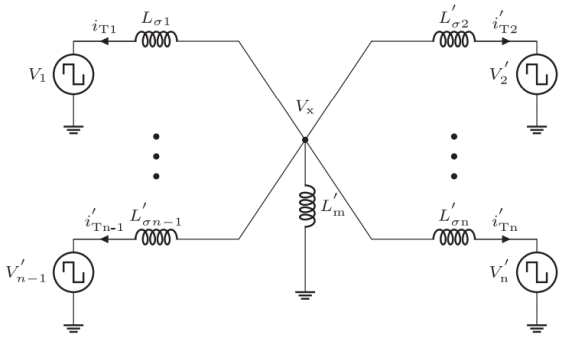


Fig. 2: Equivalent model of an MAB converter [2].

One popular method is to add a series resonant capacitor to the primary side [1]. The other effective hardware approach to decoupling the ports is to make low leakage inductances on all ports, while introducing external inductors on selected sides, thereby creating a highly asymmetrical multi-winding structure, as shown in Fig. 3. This asymmetry allows internal leakage inductance to be significantly smaller than others, and thus this port can act as the "primary side," establishing the "neutral point" voltage by regulating the primary side switching patterns, while the remaining ports function as "secondary sides." In this configuration, the power flow among secondary ports becomes naturally decoupled.

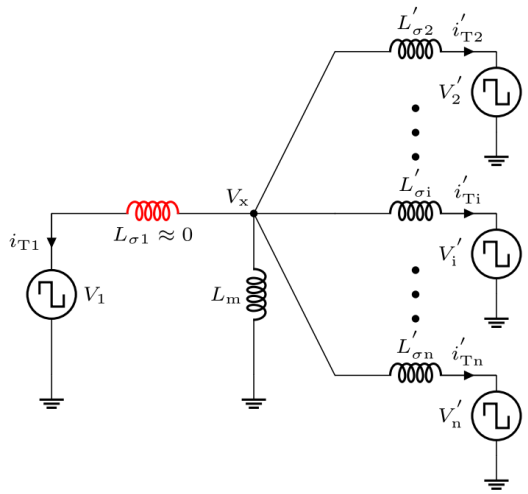


Fig. 3: Equivalent circuit when adding external inductors on secondary sides and making the primary side leakage inductance negligible [2].

For this configuration described above, the power-flow decoupling is intuitive. With $L_{\sigma 1} \approx 0$, the neutral point V_x can be rigidly determined by the switching patterns of the primary side H-bridge, yielding:

$$V_x \approx V_1.$$

With this mechanism, all secondary H-bridges can achieve decentralized power-flow control by adjusting its only phase shift with respect to primary side H-bridge, analogous to the standard operating principle of a conventional DAB converter. Therefore, by applying the basic single-phase-shift (SPS) control to each secondary side independently, the power flow from the primary side to the secondary #i side can be obtained that

$$P_i = \frac{V_1 V_i'}{2\pi L_{eq}} \phi_i \left(1 - \frac{\phi_i}{\pi}\right) T.$$

Although the incorporation of additional components increases the overall converter complexity, potentially enlarging its physical size and reducing power density, the ideal of designing a highly asymmetric transformer remains highly effective in achieving power-flow decoupling. Motivated by this observation, the proposed project investigates a new configuration in which an integrated asymmetrical multi-winding transformer is intentionally designed to realize the desired decoupled power

flow without relying on extra peripheral components, thereby making the converter more compact.

It is, however, widely recognized that multi-winding transformers are difficult to model and design due to the inherent complexity of their magnetic coupling interactions. As a result, simple equivalent circuits such as the one shown in Fig. 2 often fail to capture the full behavior of the realistic multi-winding transformer. An effective modeling approach introduced in [4] is the Extended Cantilever Model (ECM), which represents multi-winding transformers using a compact set of parameters, while still providing a necessary-and-sufficient circuit description.

However, the recommended experimental procedure for extracting the ECM parameters requires short circuit measurements, which are often unavailable in some Finite-Element Modeling (FEM) simulations that focus primarily on magnetic-field behavior rather than integrated with circuit-level behavior. Therefore, this study will instead pursue a physics-based modeling approach to derive the required parameters directly from the electrical and electromagnetic properties of the transformer.

Proposed Asymmetrical Transformer Design

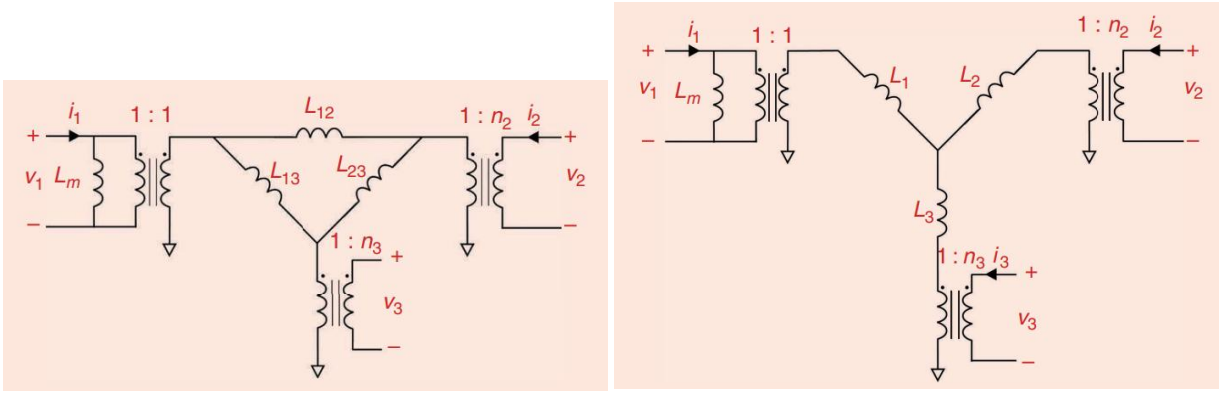


Fig. 4: Three-winding ECM, in either Delta-connection or Y-connection [4].

For this case study of triple-active-bridge (TAB), the expected three-winding ECM is shown on Fig. 4. For the three-winding ECM, the neutral point of Y-connection in the cantilever model can be written by superposition theory as

$$V_n = V_1 \frac{L_2||L_3}{L_2||L_3 + L_1} + \frac{V_2}{n_2}\frac{L_1||L_3}{L_1||L_3 + L_2} + \frac{V_3}{n_3}\frac{L_1||L_2}{L_1||L_2 + L_3}.$$

With $L_1 \ll L_2, L_3$, the above equation can be reduced to a simple representation that $V_n \approx V_1$, which aligns with the mechanism as demonstrated in the previous research of decoupled power flow [2].

Enlightened by some asymmetrical design examples for two-winding transformers, e.g. in KPVS textbook and [3], a two-winding integrated transformer design can achieve high asymmetricity by intentionally making an asymmetrical magnetic core structure, thus resulting primary side leakage being significantly smaller than the secondary side leakage inductances, to obtain the basic two-winding cantilever model. This intuition can be adopted to multi-winding cases, that asymmetrical magnetic core design can result in asymmetrical electrical equivalent structure.

Understanding these intuitions, the three-winding transformer should be designed such that the primary winding is strongly coupled to both secondary windings, while the coupling between the two secondary windings remains relatively weak. One practical approach is to use an E-type core in which the two secondary windings are placed on the outer legs, each with a relatively large air gap, while the primary winding is positioned on the center leg with a much smaller air gap.

The objective of the asymmetrical structure in the proposed structure is as follows: (1) it achieves large magnetizing inductance, (2) it achieves low leakage inductance on the primary side, and (3) it intentionally creates asymmetry among the interleaving inductances. Intuitively, in the Delta-connected ECM, if the inductances associated with the primary side (L_{12} and L_{13}) are made significantly smaller than the inductance between the two secondary sides (L_{23}), then an asymmetric inductance distribution will exist in the corresponding Y-connected ECM. In particular, this results in L_1 being much smaller than both L_2 and L_3, which aligns with the desired decoupling behavior for power-flow control.

In conclusion, this configuration, illustrated in Fig. 5, naturally produces strong primary-to-secondary coupling and weaker secondary-to-secondary coupling, thereby supporting the desired asymmetrical inductance characteristics. In the following steps, the FEM tool is used to analyze the parameters in the ECM of the proposed structure.

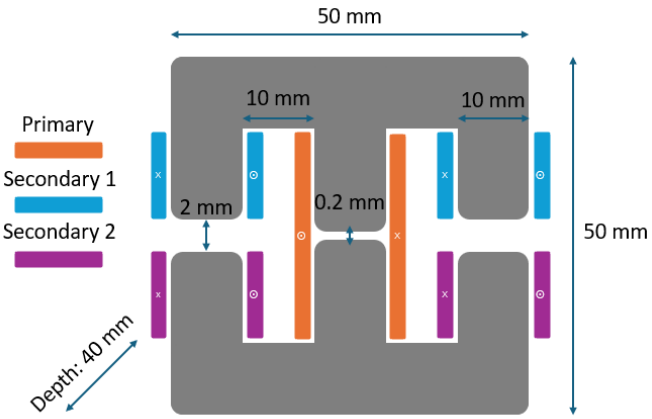


Fig. 5: Proposed three-winding structure in 2D view, with its key dimensions. The color orange, blue and purple represent the primary winding, secondary 1 winding and secondary 2 winding respectively.

As discussed in the previous section, performing short-circuit tests within FEM simulations is generally tricky. Therefore, in this study the complete inductance matrix is instead directly measured from the open-circuit solutions and electromagnetic field solutions. For a three-winding transformer, ignoring the parasitic resistance of the wirings, the inductance matrix can be expressed as:

$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = j\omega \begin{bmatrix} l_{11} & l_{12} & l_{13} \\ l_{12} & l_{22} & l_{23} \\ l_{13} & l_{23} & l_{33} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix},$$

$$\begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = \begin{bmatrix} l_{11} & l_{12} & l_{13} \\ l_{12} & l_{22} & l_{23} \\ l_{13} & l_{23} & l_{33} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}.$$

Note that lowercase letters are used here to distinguish the inductance notation from that used in the Delta-connected ECM representation shown in Fig. 4. To obtain the inductance matrix, an effective and appropriate FEM test procedure is to perform three independent excitation cases: in each case, one winding is driven with a specified current and the other windings are kept open circuits (no excitation). The resulting flux linkage and/or voltage amplitude in all windings in three test cases are measured. These flux linkages and/or voltage can then be used to compute the corresponding self- and mutual-inductance terms.

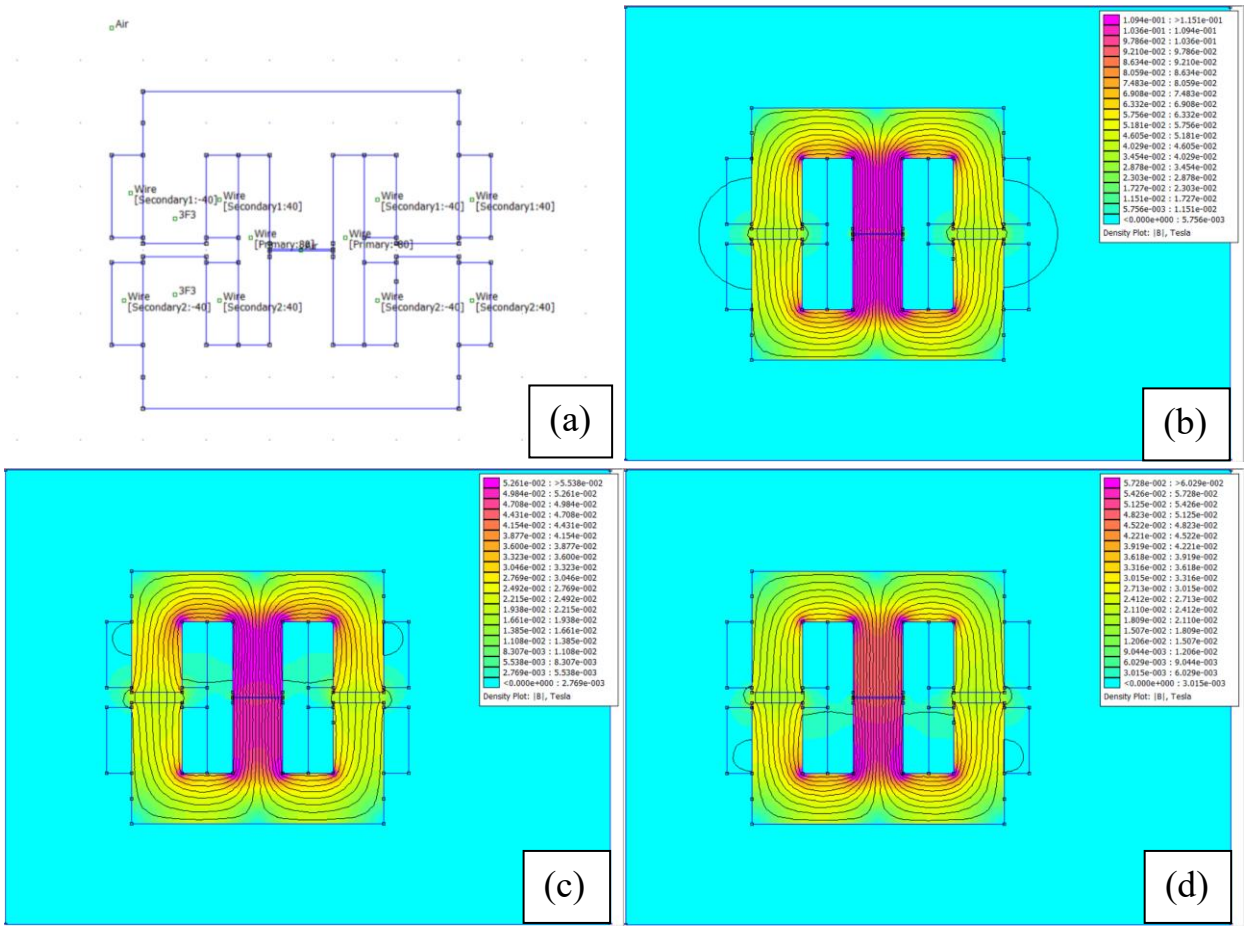


Fig. 6: (a) is the FEM modelling, (b), (c) and (d) are the results of magnetic flux density for three cases where $I_1 = 1$ A, $I_2 = 1$ A, $I_3 = 1$ A respectively.

The 2D FEM analysis is performed using the FEMM software package (free on femm.info), as illustrated in Fig. 6(a). For a high-frequency transformer intended for DCDC converter applications, the 3F3 ferrite material is selected due to its suitability for general purposes high-frequency designs, supporting operating ranges around 0.2 to 0.5 MHz and providing a saturation flux density of approximately 400 mT (ferroxcube.com). In this demonstration, each of the three windings is configured with 80 turns and modeled using Litz wire to accommodate high-frequency currents. The actual conductor size, number of turns, and thermal considerations can be scaled according to specific design requirements; these aspects are not the focus of this study, as the aim here is primarily to illustrate the viability of the proposed transformer concept.

The three FEM test cases are excited by 1 A current, and the results are shown in Fig. 6(b)-(d). From the built-in parameter analysis block in FEMM, the information of the inductances can be obtained and computed in MATLAB, and summarized into the following table, and this inductance matrix can be intuitively visualized by heatmap in Fig. 7.

Table 1: Inductance matrix (unit: μH)

	Primary	Secondary 1	Secondary 2
Primary	3673	1697	1697
Secondary 1	1697	948	785
Secondary 2	1697	785	948

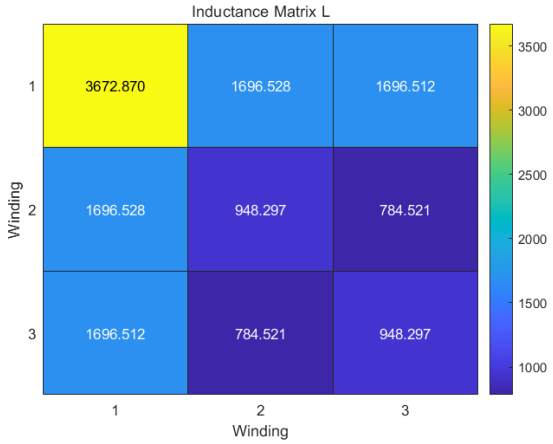


Fig. 7: Inductance matrix visualization of proposed transformer.

Thus, according to the parameter mapping described in [4], the parameters in the three-winding Y-connected ECM can be derived that:

$$L_m = l_{11} = 3673\ \mu\text{H},$$

$$n_2 = \frac{l_{12}}{l_{11}} = 0.462,$$

$$n_3 = \frac{l_{13}}{l_{11}} = 0.462,$$

$$L_1 = \frac{L_{23}L_{11}^2}{L_{12}L_{13}} - L_{11} = 4.16\ \mu\text{H},$$

$$L_2 = \frac{L_{22}L_{11}^2}{L_{12}^2} - \frac{L_{23}L_{11}^2}{L_{12}L_{13}} = 767.57\ \mu\text{H},$$

$$L_3 = \frac{L_{33}L_{11}^2}{L_{13}^2} - \frac{L_{23}L_{11}^2}{L_{12}L_{13}} = 767.66\ \mu\text{H}.$$

The results obtained from the Y-connected ECM can be converted to the Delta-connected ECM through a standard Y-Delta transformation. In the transformed model, it becomes evident that L_{23} is significantly larger than L_{12} and L_{13}. These outcomes, especially the values of L_1, L_2, and L_3 (or equivalently L_{12}, L_{13}, and L_{23}), demonstrate the strong effectiveness of the proposed method. Specifically, the primary winding exhibits strong coupling with each of the two secondary windings, while the coupling between the secondary windings remains intentionally weak, reflected by the much larger equivalent inductance. This inductance profile directly supports the desired decoupled power-flow behavior.

It is noteworthy that the proposed structure is intended primarily to demonstrate the idea of an asymmetrical transformer for decoupled TAB operation. It is not using the commercial product (E-type core). Further optimization would be required to adapt the concept to commercially available transformer platforms or to design a customized transformer suitable for production. Additionally, power rating considerations, including losses, thermal, have not been fully scaled or optimized in this preliminary study. Therefore, in the following analysis, a relatively low power level is selected to have a large design margin.

MATLAB Validation

To validate the proposed three-winding asymmetric transformer design, a MATLAB/Simulink simulation is conducted using the setup shown in Fig. 8. For simplicity, an open-loop control strategy with fixed phase-shift SPS is employed. This allows us to verify that phase-shift commands applied to each secondary H-bridge do not influence the power flow of the other secondaries, thereby confirming decentralized and decoupled control behavior.

In the test scenario, the phase shift between secondary 1 and the primary H-bridge is set to $\phi_2 = 0.2\pi$, while that between secondary 2 and the primary H-bridge is set

to $\phi_3 = 0.1\pi$. Furthermore, secondary 2 is activated after secondary 1 has already been operating, allowing observation of the dynamic interaction between the two secondaries. Under ideal conditions with fully decoupled power flow, the resulting transferred powers should satisfy:

$$P_2 = \frac{V_1 V_2}{2\pi(L_1 + L_2)n_2}\phi_2\left(1 - \frac{\phi_2}{\pi}\right)T \approx 22\ \text{W},$$

$$P_3 = \frac{V_1 V_3}{2\pi(L_1 + L_3)n_3}\phi_3\left(1 - \frac{\phi_3}{\pi}\right)T \approx 13\ \text{W},$$

which leads to the mean value of periodic current of secondary H-bridges that

$$I_{2,mean} \approx 0.22\ \text{A},$$

$$I_{3,mean} \approx 0.13\ \text{A}.$$

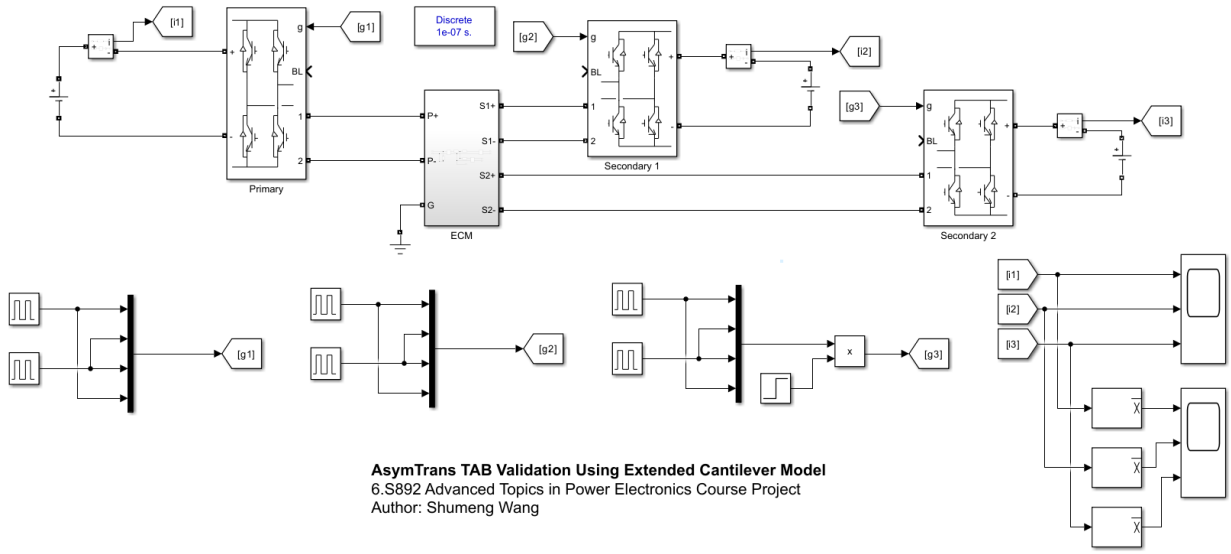


Fig. 8: Simulink simulation setup.

The simulation results are presented in Fig. 9. When secondary #2 H-bridge is enabled, the transient response is smooth, and its current quickly settles to the expected value. Importantly, secondary #1 exhibits no noticeable transient disturbance and no measurable deviation in its steady-state operating point. The steady-state currents of all three ports match the theoretical values derived earlier, thereby confirming that the proposed asymmetrical transformer structure effectively enables decoupled power flow among the secondary ports.

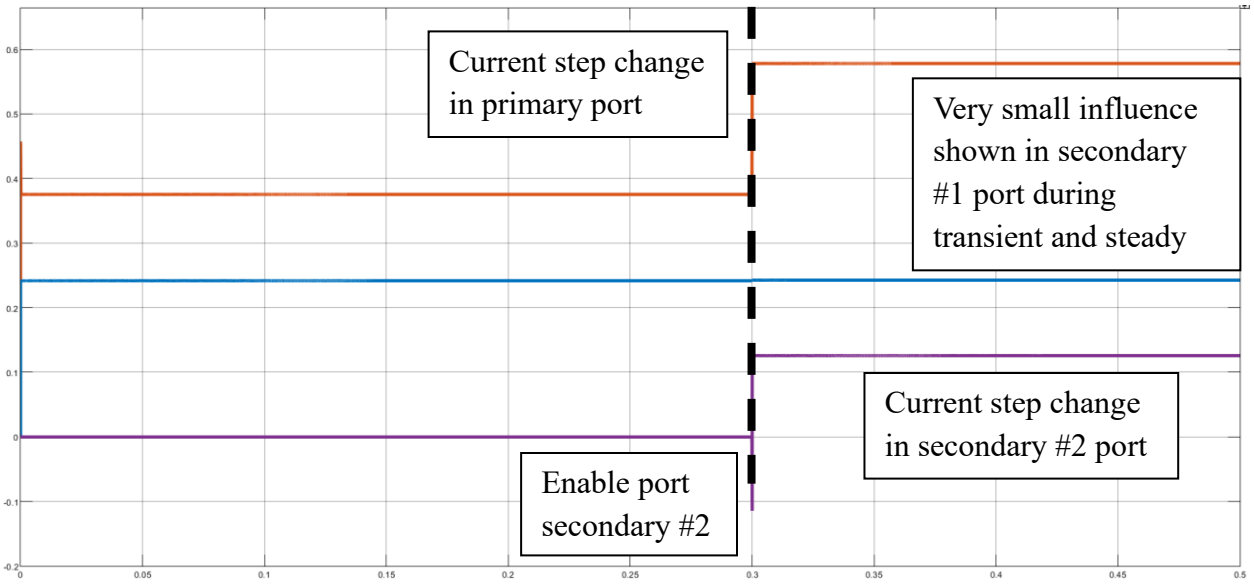


Fig. 9: Simulation results of TAB. The color orange, blue and purple represent the periodic mean value of currents of DC sides of three bridges.

Future Work

The effectiveness of the 2D three-winding transformer structure does not necessarily extend to designs with a higher number of windings. As illustrated in Fig. 10 for the four-winding case, no "neutral point" naturally emerges in the magnetic model, and thus the earlier design strategy cannot be directly generalized. Consequently, alternative approaches must be considered to achieve an integrated transformer that enables decoupled power flow.

For example, one intuitive idea is to design the transformer such that all leakage inductances associated with the primary side (i.e., L_{12}, L_{13}, and L_{14}) are significantly smaller than the leakage inductances among the secondary windings (i.e., L_{23}, L_{24}, L_{34}). Under this condition, the magnetic coupling between secondary windings becomes weak and can be approximated as an open circuit. As long as L_{12}, L_{13}, and L_{14} maintain the desired values corresponding to the required power-flow scaling factors, this configuration could in principle achieve decoupled power flow similar to the three-winding case.

However, realizing such an inductance distribution in practice is challenging. Achieving the required strong primary-to-secondary coupling while simultaneously

suppressing secondary-to-secondary coupling may be difficult from various aspects. Therefore, this concept requires further investigation to determine whether it is physically realistic and practical from a design perspective.

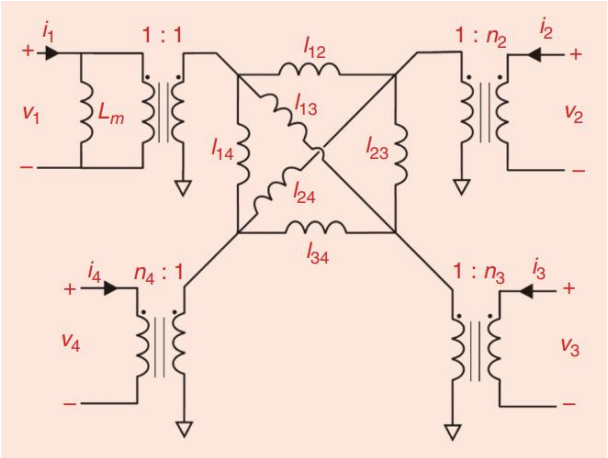


Fig. 10: Four-winding ECM [4].

Open Source

The design files and MATLAB code and simulation files are available on GitHub repository: https://github.com/swangneu/AsymTrans

References

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